

Functions & Graphs

ANSWER KEY



Function transformations

- 1 Let $f(x) = x^2$ and $g(x) = -3f(x+2) - 4 = -3(x+2)^2 - 4$. Find $g(0)$, and describe the transformation from f to g in words.

$g(0) = -3(2)^2 - 4 = -3(4) - 4 = -16$. g is f shifted left 2, reflected over the x -axis and stretched vertically by a factor of 3, then shifted down 4.

- 2 Using the same $g(x) = -3(x+2)^2 - 4$, find $g(-2)$, and explain why this is the vertex of g 's graph.

$g(-2) = -3(0)^2 - 4 = -4$. Since $f(x) = x^2$ has its vertex (minimum) at $x = 0$, and g shifts f left by 2, the transformed vertex sits at $x = -2$.

Even and odd functions

- 3 Classify $f(x) = x^4 + 2x^2 - 5$ as even, odd, or neither.

Only even-degree terms appear (x^4 , x^2 , and the constant), so by Theorem 3.1 f is even. Check: $f(2) = 16 + 8 - 5 = 19$ and $f(-2) = 16 + 8 - 5 = 19$ ✓.

- 4 Classify $h(x) = 2x^5 - 3x$ as even, odd, or neither.

Only odd-degree terms appear (x^5 , x), so by Theorem 3.1 h is odd. Check: $h(1) = 2 - 3 = -1$ and $h(-1) = -2 + 3 = 1 = -h(1)$ ✓.

- 5 Explain why $k(x) = x^2 + x$ is neither even nor odd.

k mixes an even-degree term (x^2) with an odd-degree term (x), so Theorem 3.1 doesn't apply to the whole function. Checking directly: $k(1) = 2$ and $k(-1) = 0$, which is neither equal to $k(1)$ (even) nor $-k(1) = -2$ (odd).

Domain restrictions

- 6 Find the domain of $f(x) = \sqrt{2x+6}$.

Need $2x+6 \geq 0 \Rightarrow x \geq -3$. Domain: $x \geq -3$.

- 7 Find the domain of $g(x) = \frac{1}{x^2-25}$.

Need $x^2 - 25 \neq 0 \Rightarrow x \neq \pm 5$. Domain: all reals except 5 and -5.

Piecewise functions

- 8 Let $p(x) = 2x + 3$ for $x < 1$, and $p(x) = x^2 - 1$ for $x \geq 1$. Find $p(-2)$ and $p(3)$.

Since $-2 < 1$, use the first piece: $p(-2) = 2(-2) + 3 = -1$. Since $3 \geq 1$, use the second piece: $p(3) = 3^2 - 1 = 8$.

Multiple choice — easy

- 9 Classify $f(x) = 2x^3 + 5$ as even, odd, or neither.

- a. even
- b. both even and odd
- c. neither ✓
- d. odd

$f(-x) = 2(-x)^3 + 5$, which equals neither $f(x)$ nor $-f(x)$, so f is neither even nor odd.

- 10 Classify $f(x) = 3x^3 + x$ as even, odd, or neither.

- a. both even and odd
- b. even
- c. neither
- d. odd ✓

$f(-x) = 3(-x)^3 + (-x) = -f(x)$, so f is odd.

- 11 Classify $f(x) = -3x^5 + 6x$ as even, odd, or neither.

- a. both even and odd
- b. even
- c. neither
- d. odd ✓

$f(-x) = -3(-x)^5 + 6(-x) = -f(x)$, so f is odd.

12 Classify $f(x) = 3x^2 + 2$ as even, odd, or neither.

a. both even and odd

b. neither

c. even ✓

d. odd

$f(-x) = 3(-x)^2 + 2 = f(x)$, so f is even.

13 Classify $f(x) = -x^2 + 6$ as even, odd, or neither.

a. both even and odd

b. neither

c. odd

d. even ✓

$f(-x) = -(-x)^2 + 6 = f(x)$, so f is even.

14 Classify $f(x) = 2x^5 + 2x$ as even, odd, or neither.

a. odd ✓

b. even

c. neither

d. both even and odd

$f(-x) = 2(-x)^5 + 2(-x) = -f(x)$, so f is odd.

15 Classify $f(x) = 3x^5 + 6x$ as even, odd, or neither.

a. even

b. neither

c. odd ✓

d. both even and odd

$f(-x) = 3(-x)^5 + 6(-x) = -f(x)$, so f is odd.

16 Classify $f(x) = -2x^2 + 7$ as even, odd, or neither.

- a. odd
- b. both even and odd
- c. neither
- d. even ✓

$f(-x) = -2(-x)^2 + 7 = f(x)$, so f is even.

17 Classify $f(x) = 3x^3 + 5x$ as even, odd, or neither.

- a. even
- b. odd ✓
- c. both even and odd
- d. neither

$f(-x) = 3(-x)^3 + 5(-x) = -f(x)$, so f is odd.

18 Classify $f(x) = -2x^3 + 8$ as even, odd, or neither.

- a. neither ✓
- b. odd
- c. both even and odd
- d. even

$f(-x) = -2(-x)^3 + 8$, which equals neither $f(x)$ nor $-f(x)$, so f is neither even nor odd.

19 Classify $f(x) = 2x^2 + 4x$ as even, odd, or neither.

- a. both even and odd
- b. odd
- c. **neither ✓**
- d. even

$f(-x) = 2(-x)^2 + 4(-x)$, which equals neither $f(x)$ nor $-f(x)$, so f is neither even nor odd.

20 Classify $f(x) = -3x^3 + 4x$ as even, odd, or neither.

- a. even
- b. neither
- c. both even and odd
- d. **odd ✓**

$f(-x) = -3(-x)^3 + 4(-x) = -f(x)$, so f is odd.

21 Classify $f(x) = -3x^3 + 4$ as even, odd, or neither.

- a. odd
- b. both even and odd
- c. **neither ✓**
- d. even

$f(-x) = -3(-x)^3 + 4$, which equals neither $f(x)$ nor $-f(x)$, so f is neither even nor odd.

22 Classify $f(x) = -x^4 + 8x^2$ as even, odd, or neither.

- a. odd
- b. both even and odd
- c. **even ✓**
- d. neither

$f(-x) = -(-x)^4 + 8(-x)^2 = f(x)$, so f is even.

23 Classify $f(x) = -2x^4 + 8x^2$ as even, odd, or neither.

- a. odd
- b. neither
- c. even ✓
- d. both even and odd

$f(-x) = -2(-x)^4 + 8(-x)^2 = f(x)$, so f is even.

Multiple choice — medium

24 Find the domain of $f(x) = \sqrt{5x - 35}$.

- a. $x \geq 9$
- b. $x \geq 7$ ✓
- c. $x \geq 5$
- d. $x \leq 7$

Need $5x - 35 \geq 0 \Rightarrow x \geq 7$. Domain: $x \geq 7$.

25 Find the domain of $f(x) = \sqrt{x + 1}$.

- a. $x \geq 1$
- b. $x \leq -1$
- c. $x \geq -1$ ✓
- d. $x \geq -3$

Need $x + 1 \geq 0 \Rightarrow x \geq -1$. Domain: $x \geq -1$.

26 Find the domain of $f(x) = \sqrt{x-1}$.

a. $x \geq -1$

b. $x \geq 1$ ✓

c. $x \leq 1$

d. $x \geq 3$

Need $x - 1 \geq 0 \Rightarrow x \geq 1$. Domain: $x \geq 1$.

27 Find the domain of $f(x) = \sqrt{4x-24}$.

a. $x \geq 4$

b. $x \leq 6$

c. $x \geq 8$

d. $x \geq 6$ ✓

Need $4x - 24 \geq 0 \Rightarrow x \geq 6$. Domain: $x \geq 6$.

28 Find the domain of $f(x) = \sqrt{4x-20}$.

a. $x \geq 7$

b. $x \leq 5$

c. $x \geq 5$ ✓

d. $x \geq 3$

Need $4x - 20 \geq 0 \Rightarrow x \geq 5$. Domain: $x \geq 5$.

29 Find the domain of $f(x) = \sqrt{x+3}$.

a. $x \leq -3$

b. $x \geq -5$

c. $x \geq -3$ ✓

d. $x \geq -1$

Need $x + 3 \geq 0 \Rightarrow x \geq -3$. Domain: $x \geq -3$.

30 Find the domain of $f(x) = \sqrt{-x + 5}$.

a. $x \geq 5$

b. $x \leq 5$ ✓

c. $x \leq 7$

d. $x \leq 3$

Need $-x + 5 \geq 0 \Rightarrow x \leq 5$. Domain: $x \leq 5$.

31 Find the domain of $f(x) = \sqrt{5x + 25}$.

a. $x \geq -5$ ✓

b. $x \geq -7$

c. $x \leq -5$

d. $x \geq -3$

Need $5x + 25 \geq 0 \Rightarrow x \geq -5$. Domain: $x \geq -5$.

32 Find the domain of $f(x) = \sqrt{4x - 28}$.

a. $x \geq 9$

b. $x \geq 7$ ✓

c. $x \leq 7$

d. $x \geq 5$

Need $4x - 28 \geq 0 \Rightarrow x \geq 7$. Domain: $x \geq 7$.

33 Find the domain of $f(x) = \sqrt{5x - 20}$.

- a. $x \geq 6$
- b. $x \leq 4$
- c. $x \geq 2$
- d. $x \geq 4$ ✓

Need $5x - 20 \geq 0 \Rightarrow x \geq 4$. Domain: $x \geq 4$.

34 Find the domain of $f(x) = \sqrt{5x}$.

- a. $x \geq 0$ ✓
- b. $x \geq 2$
- c. $x \leq 0$
- d. $x \geq -2$

Need $5x \geq 0 \Rightarrow x \geq 0$. Domain: $x \geq 0$.

35 Find the domain of $f(x) = \sqrt{-3x - 6}$.

- a. $x \leq -4$
- b. $x \leq 0$
- c. $x \leq -2$ ✓
- d. $x \geq -2$

Need $-3x - 6 \leq 0 \Rightarrow x \leq -2$. Domain: $x \leq -2$.

36 Find the domain of $f(x) = \sqrt{-2x + 12}$.

- a. $x \leq 6$ ✓
- b. $x \leq 8$
- c. $x \leq 4$
- d. $x \geq 6$

Need $-2x + 12 \leq 0 \Rightarrow x \leq 6$. Domain: $x \leq 6$.

37 Find the domain of $f(x) = \sqrt{5x + 40}$.

a. $x \geq -10$

b. $x \geq -6$

c. $x \geq -8$ ✓

d. $x \leq -8$

Need $5x + 40 \geq 0 \Rightarrow x \geq -8$. Domain: $x \geq -8$.

38 Find the domain of $f(x) = \sqrt{-3x - 27}$.

a. $x \leq -11$

b. $x \geq -9$

c. $x \leq -7$

d. $x \leq -9$ ✓

Need $-3x - 27 \leq 0 \Rightarrow x \leq -9$. Domain: $x \leq -9$.

Multiple choice — hard

39 Let $p(x) = 2x + 1$ for $x < -2$, and $p(x) = 3x - 6$ for $x \geq -2$. Find $p(-6)$ and $p(-2)$.

a. $-12, -11$

b. $-11, -14$

c. $-11, -12$ ✓

d. $-9, -12$

Since $-6 < -2$, use the first piece: $p(-6) = 2(-6) + 1 = -11$. Since $-2 \geq -2$, use the second piece: $p(-2) = 3(-2) - 6 = -12$.

40 Let $p(x) = 3x - 6$ for $x < 3$, and $p(x) = -2x + 3$ for $x \geq 3$. Find $p(-2)$ and $p(7)$.

a. $-12, -11$ ✓

b. $-12, -13$

c. $-10, -11$

d. $-11, -12$

Since $-2 < 3$, use the first piece: $p(-2) = 3(-2) - 6 = -12$. Since $7 \geq 3$, use the second piece: $p(7) = -2(7) + 3 = -11$.

41 Let $p(x) = 3x - 6$ for $x < 2$, and $p(x) = 2x - 4$ for $x \geq 2$. Find $p(1)$ and $p(6)$.

a. $8, -3$

b. $-3, 6$

c. $-3, 8$ ✓

d. $-1, 8$

Since $1 < 2$, use the first piece: $p(1) = 3(1) - 6 = -3$. Since $6 \geq 2$, use the second piece: $p(6) = 2(6) - 4 = 8$.

42 Let $p(x) = x + 6$ for $x < 0$, and $p(x) = -5x + 6$ for $x \geq 0$. Find $p(-5)$ and $p(5)$.

a. $1, -19$ ✓

b. $-19, 1$

c. $3, -19$

d. $1, -21$

Since $-5 < 0$, use the first piece: $p(-5) = (-5) + 6 = 1$. Since $5 \geq 0$, use the second piece: $p(5) = -5(5) + 6 = -19$.

43 Let $p(x) = 4x - 7$ for $x < 5$, and $p(x) = -x + 1$ for $x \geq 5$. Find $p(3)$ and $p(5)$.

a. 5, -6

b. 5, -4 ✓

c. -4, 5

d. 7, -4

Since $3 < 5$, use the first piece: $p(3) = 4(3) - 7 = 5$. Since $5 \geq 5$, use the second piece: $p(5) = -(5) + 1 = -4$.

44 Let $f(x) = x^2$ and $g(x) = f(x + 4) - 3 = (x + 4)^2 - 3$. Find $g(-5)$.

a. -6

b. 4

c. 2

d. -2 ✓

$g(-5) = (-1)^2 - 3 = (1) - 3 = -2$.

45 Let $f(x) = x^2$ and $g(x) = 4f(x - 3) - 1 = 4(x - 3)^2 - 1$. Find $g(4)$.

a. 7

b. -1

c. -3

d. 3 ✓

$g(4) = 4(1)^2 - 1 = 4(1) - 1 = 3$.

46 Let $f(x) = x^2$ and $g(x) = -2f(x - 4) - 6 = -2(x - 4)^2 - 6$. Find $g(3)$.

a. 8

b. -12

c. -8 ✓

d. -4

$$g(3) = -2(-1)^2 - 6 = -2(1) - 6 = -8.$$

47 Let $f(x) = x^2$ and $g(x) = -f(x - 2) + 8 = -(x - 2)^2 + 8$. Find $g(-8)$.

a. -96

b. -92 ✓

c. -88

d. 92

$$g(-8) = -(-10)^2 + 8 = -(100) + 8 = -92.$$

48 Let $f(x) = x^2$ and $g(x) = 4f(x + 2) + 6 = 4(x + 2)^2 + 6$. Find $g(-8)$.

a. 150 ✓

b. 146

c. -150

d. 154

$$g(-8) = 4(-6)^2 + 6 = 4(36) + 6 = 150.$$

49 Find the domain of $g(x) = \frac{1}{x^2 - 64}$.

a. 8, -10

b. -8, 8

c. 10, -8

d. 8, -8 ✓

Need $x^2 - 64 \neq 0 \Rightarrow x \neq \pm 8$. Domain: all reals except 8 and -8.

50 Find the domain of $g(x) = \frac{1}{x^2 - 144}$.

a. 14, -12

b. 12, -12 ✓

c. -12, 12

d. 12, -14

Need $x^2 - 144 \neq 0 \Rightarrow x \neq \pm 12$. Domain: all reals except 12 and -12.

51 Find the domain of $g(x) = \frac{1}{x^2 - 16}$.

a. 4, -4 ✓

b. 6, -4

c. -4, 4

d. 4, -6

Need $x^2 - 16 \neq 0 \Rightarrow x \neq \pm 4$. Domain: all reals except 4 and -4.

52 Find the domain of $g(x) = \frac{1}{x^2 - 49}$.

a. 9, -7

b. 7, -9

c. 7, -7 ✓

d. -7, 7

Need $x^2 - 49 \neq 0 \Rightarrow x \neq \pm 7$. Domain: all reals except 7 and -7.

53 Find the domain of $g(x) = \frac{1}{x^2 - 9}$.

a. 5, -3

b. 3, -3 ✓

c. 3, -5

d. -3, 3

Need $x^2 - 9 \neq 0 \Rightarrow x \neq \pm 3$. Domain: all reals except 3 and -3.